「Preliminary.

「Prop. Let $\varphi : R \to S$ be a non-trivial Ring Homomorphism, where R, S have identity, $1_R, 1_S$ respectively.
Then, either φ preserves identity or φ maps 1_R to zero divisor of S.

Proof. Since $1_R \cdot 1_R = 1_R$ and φ preserves multiplication,

$$\varphi(1_R) = \varphi(1_R \cdot 1_R) = \varphi(1_R) \cdot \varphi(1_R) \text{ implies } \varphi(1_R)(1_S - \varphi(1_R)) = 0.$$

Therefore, if $\varphi(1_R) \neq 1_S$, then $\varphi(1_R)$ must be zero divisor or zero.
$\varphi(1_R) = 0$ implies for any $a \in R$, $\varphi(a) = \varphi(a) \cdot \varphi(1_R) = \varphi(a) \cdot 0 = 0$,
So it cannot be happen, by the assumption $\boxtimes$

Def.

Let $\varphi : F \to F'$ be a field homomorphism.

Define $\varphi_x : F[x] \to F'[x] : \sum a_n \cdot x^n \mapsto \sum \varphi(a_n) x^n$,

is called the Natural Homomorphism